卷册检索号 SEARCH NO. OF THE VOLUME
MLHVDC-LCS-ECI-DD-CS-0509 30-B003701S-T0509

第 1 页 共 1 页
PAGE 1 OF 1

图纸文件目录（首页）

CATALOG OF DRAWING DOCUMENTS (HEAD PAGE)

拉合尔±660kV换流站 工程 施工图 阶段
±660kV Lahore Converter Station PROJECT Detail Design DESIGN STAGE

结构 专业 第 05 卷 第 09 册
Structure SPECIALITY VOLUME BOOK

卷册名称 阀冷废水排水池结构图
Volume name Structural drawings of drainage pool for valve-cooled wastewater

批 准 APPROVED BY	
审 核 REVIEWED BY	
设计/勘测 DESIGNED BY/ SURVEYED BY	付国顺
图 纸 3 张 DRAWING NOS.	打印件 0 本 PRINTED COPIES NOS.

序号 No.	图 号 DRAWING NO.	文 件 名 称 DOCUMENT TITLE	版本 EDITION	张 NOS.	本 NOS.	备注 REMARKS
1	30-B003701S-T0509-001	地下构筑物结构设计说明 Specification of structural design for underground structures	A	1		
2	30-B003701S-T0509-002	水池平面图、剖面图 Plane and cross-section of pool	A	1		
3	30-B003701S-T0509-003	水池配筋图 Pool reinforcement details	A	1		

注:

设 计 总 说 明

1、本构筑物设计依据：水工提资及地质勘测报告、国家规范规程。
2、场地-0.300相当于绝对标高193.40,±0.000相当于绝对标高193.70(EGM2008高程系统)。
如无特别说明，本工程标高均采用相对标高。
3、采用巴基斯坦UTM-43N平面坐标系，相对平面坐标以米(m)为单位，尺寸以毫米(mm)为单位。
4、根据《巴基斯坦建筑规范》(2007)，换流站属于ZONE 2A区，场地土类型为SD类。根据巴基斯坦
地面峰值加速度图，地面峰值加速度为0.14g,相当于国内地震基本烈度为7度。
5、本构筑物安全等级二级，设计使用年限50年。地下工程防水等级二级。地基基础设计等级丙级。
6、设计荷载：地面活荷载4.0kN/m²。
7、基础底应置于良好土层上，如遇不良地基请及时通知设计院。施工期间应注意采取措施防止构筑物上浮。
8、混凝土标号：fc'=28MPa，技术要求：(1)水胶比不大于0.50，(2)氯离子含量不大于0.1%，
(3)水泥型号应满足ACI-318-14表19.3.2.1中暴露类别S1对应水泥
(Cementitious materials-Types)的要求，(4)抗渗等级0.8MPa，(5)掺矿物掺合料。
混凝土垫层fc'=21MPa。钢筋Φ代表Grade 60 钢，技术要求：应满足ASTM A706, Grade 60[420]
变形钢筋或满足ACI 318-14第20.2.2.5节要求的ASTM A615, Grade 60[420]变形钢筋。
钢筋的抗拉强度实测值与屈服强度实测值的比值不应小于1.25；钢筋的屈服强度实测值与
屈服强度标准值的比值不应大于1.3，且钢筋在最大拉力下的总伸长率实测值不应小于9%。
钢筋Φ代表Grade 40，钢材Q235B，电焊条E43XX，焊缝高度Hf=h_{min}。
所有外露铁件均热镀锌防腐，焊伤处防锈漆一道，银粉漆二道。
9、粉刷：(1)水池内壁，支柱，顶板底面采用清水混凝土工艺。
(2)底板顶面，采用清水混凝土工艺。
10、受力钢筋的混凝土保护层厚度
顶板上、下均为35mm；底板上为50mm，下为75mm；钢筋混凝土梁、柱50mm。
池壁内侧为50mm，外侧为75mm。
11、水池结构及人孔等配件按国标图纸05S804有关图纸施工。
12、施工要求：
(1)水池施工，安装及验收均应遵照现行建筑施工验收规范进行。
(2)混凝土：
(a)水池混凝土浇筑时必须振捣密实，不得漏振。
(b)池壁施工缝的位置可以设在以下二处：
<1>底板与池壁连接的斜托上部；
<2>顶板与池壁连接的斜托下部。
(c)浇筑水池混凝土前应将爬梯、墙管和吊攀等预埋件按图予先埋设牢固，防止浇筑混凝土时松动。
(d)在地下构筑物侧壁内外、底板上下分别涂抹两层厚度为10~12mm的聚合物水泥防水砂浆。
(3)施工期间注意基坑排水，防止水池上浮。
(4)水池土建完成以后，覆土回填工作应在试水合格后立即回填，
沿水池四周及池顶分层均匀回填，防止超填。顶板表面覆土时要避免大力夯打。
(5)水池抹面之前先做充水试验，充水分三次，每次充三分之一水深，每次充水结束稳定二天，观察和测定
渗漏情况，扣除管道的渗漏因素，24小时渗漏率应小于2L(m d)，根据观察到的渗漏，视具体情况修补。
(6)冬季施工应按有关规范规定执行。
13、施工、安装及验收应遵照下列及相关的施工验收规范：
(1)<<混凝土结构工程施工质量验收规范>>GB50204-2011
(2)<<砌体结构工程施工质量验收规范>>GB50203-2011
(3)<<地下工程防水技术规范>>GB50108-2008
14、受拉钢筋的锚固长度 L_a(mm)：

钢筋种类	抗震等级	钢筋直径	混凝土强度fc'	
			21.0	28.0
		10	53d	52d
		13	53d	45d
		16	53d	45d
		19	53d	45d
		22	52d	45d
		25	52d	45d

注
1.任何情况下，纵向受拉钢筋的锚固长度不应小于250mm。
2.当钢筋在混凝土施工过程中易受扰动时，表中锚固长度应乘以修正系数1.1。
3.纵向受拉钢筋末端采用弯钩或机械锚固措施时，包括弯钩或锚固端头
在内的锚固长度(投影长度)可取表中数字的0.6倍。
4.纵向受压钢筋的最小锚固长度不应小于表中数字的0.7倍。

15、纵向受压钢筋的最小搭接长度=0.7L_a，且≥200mm。
纵向受拉钢筋的搭接长度 L_l　　$L_l=L_a\times$ 1.2 (≤25%)；1.4 (50%)；1.6 (100%)
括号内数字为纵向钢筋搭接接头面积百分率
16、套用图集及图纸：

水管吊架详图	国家建筑标准设计图集05S804 P172
不锈钢盖板详图	国家建筑标准设计图集05S804 P174
套管洞加固详图	国家建筑标准设计图集05S804 P185
总说明	国家建筑标准设计图集05S804 P6-9

Design General Specification

1. Design basis of this structure: Hydraulic capital and geological survey report, the national code of practice.
2. The site level of -0.300 is equal to the absolute elevation of193.40,The site level of ±0.000 is equal to the absolute elevation of193.70(geodestic datum system EGM2008).
If not specified,the elevations of the project are adopt to the relative elevation.
3. Using Pakistan UTM-43N plane coordinate system,the relative plane coordinates are in meters (m) and the dimensions are in millimeters (mm).
4. According to the "Pakistan Building Code" (2007), the converter station belongs to zone ZONE 2A and the site soil type is SD.According to Pakistan's ground peak acceleration map, the ground peak acceleration is 0.14g, equivalent to the basic earthquake intensity of 7 degrees.
5. The structure safety level: second-class, Design life : 50 years, Underground waterproof rating: second-class, Foundation design level : C.
6. Design load: Floor live loads: 4kN/m².
7. The foundation should be placed in good stratum, in case of poor condition, please contact ECEPDI before the construction work started.Measures should be taken to prevent the building from floating during construction.
8. Strength grading of concrete: fc'=28MPa, technical requirements: (1)the water binder ratio is not greater than 0.50, (2)the content of chloride ion is not more than 0.1%, (3)the cement shall meet the requirements of the exposure category S1 in the Table 19.3.2.1 of ACI-318-14, corresponding to the cement (Cementitious, materials-Types), (4)the impermeability grade is 0.8MPa, (5)the mineral admixture is added.
The strength of concrete used in cushion is fc'=21Mpa.
Reinforcement Φ: ASTM A615 Grade 60, fy = 420MPa., technical requirements: (1)It shall meet the requirements of ASTM A706, Grade 60[420] deformed steel bars or meeting the requirements of section ACI 20.2.2.5 318-14.The ratio between measured tensile strength and measured value of yield strength should not be less than 1.25;The ratio between the measured yield strength and the standard value of yield strength should not exceed 1.3; The measured value of the total elongation at the maximum tensile force should not be less than 9%.
Reinforcement Φ: ASTM A615 Grade 40,,Steel: Q235B, welding rod: E43XX, weld height: Hf=h_{min}.
All exposed iron parts are hot-dip galvanized anti-corrosion,welding wound apply a antirust paint and two silver powder.
9. Paint: (1)Fair-faced concrete technology was adopted on the Pool walls, pillars, roof bottom.
(2)Fair-faced concrete technology was adopted on Floor top.
10. The thickness of the protective layer of the reinforcement of different parts:
The top and bottom of the roof were 30mm; The top of the floor are 50mm,The bottom of the floor are 75mm ;
The reinforced concrete beam and column are 50mm; The inside of pool walls are 50mm , outside are 75mm.
11. Pool structure and manhole and other accessories were built according to national standard drawings 05S804. and other related contents of the atlas implementation.
12. Construction requirements:
(1) Construction, installation and acceptance of ponds shall be conducted in accordance with the current code of acceptance of construction.
(2) Concrete:
(a)Concrete pouring pond must be vibrated dense.No leak vibration.
(b)Pool wall construction joints can be located in the following two locations:
<1>Skeleton upper support connected to the upper wall of the spire;
<2>Skeleton lower support connected to the upper wall of the spire;
(c)The stairs, wall pipe , lifting and other embedded parts should be firmly embedded according to drawings before pouring pool concrete , To prevent embedded parts loosing when pouring pool concrete.
(d)Apply two layers of polymer cement waterproof mortar with a thickness of 10 ~ 12mm inside and outside the underground structure.
(3)Pay attention to excavation drainage during construction,in case to prevent the pool from floating.
(4)After the completion of the pool construction,Cover soil backfill should be tested immediately after the water backfill. Cover soil along the pool around the pool top and uniform layered backfill.Prevent overfill.Avoid aggressive ramming when top surface covered with soil.
(5)Make water test before filling the pool,Water filling three times,1/3 depth of water each time,keep two days after filling, Observe and measure leakage.Deducting the leakage of the pipeline, 24-hour leakage rate should be less than 2L(m d). According to the observed leakage,patching depending on the case.
(6)Winter construction should be implemented according to the relevant regulations.
13. Construction, installation and acceptance shall comply with the following and related construction acceptance specifications:
(1)<<Code for quality acceptance of concrete structure construction>>GB50204-2011
(2)<<Code for acceptance of constructional quality of masonry structures>>GB50203-2011
(3)<<Technical code for waterproofing of underground works>>GB50108-2008
14. The anchorage length of longitudinal tensile reinforcement:

Type of bar	Class of seismic measure	Diameter of bar	Concrete strength fc'	
			21.0	28.0
		10	53d	52d
		13	53d	45d
		16	53d	45d
		19	53d	45d
		22	52d	45d
		25	52d	45d

Notes:
1. In any case, the anchorage length of the longitudinal tensile bar shall not be less than 250mm .
2. When the end of longitudinal tensile bar is bent or mechanically anchored, The anchor length (projection length) including the hook or anchor end,is 0.6 times the figure in the table.
3. The minimum anchorage length of longitudinally compression bar shall not be less than 0.7 times the figures in the table.

15. When the longitudinal compressive bar is overlapped, the compressive lap length shall not be less than 0.7 times of the length of the longitudinal tensile bar, and shall not be less than 200mm in any case.
The lashing lap length of longitudinal tensile bars L_l　　$L_l=L_a\times$ 1.2 (≤25%)；1.4 (50%)；1.6 (100%)
The numbers in brackets are the percentage of overlap area of the bars in the same connecting section
16. Related atlas and drawings:

Plumbing details	National Building Standard Design Gallery 05S804 P172
stainless steel cover details	National Building Standard Design Gallery 05S804 P174
Cased hole reinforcement details	National Building Standard Design Gallery 05S804 P185
General Information	National Building Standard Design Gallery 05S804 P6-9

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程
±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC EQUIPMENT TECHNOLOGY COLTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站
±660kV Lahore Converter Station

设计阶段 PHASE：施工图 Detail Design

M L H V D C - L C S - E C I - D D - C S - 0 5 0 9 - 0 0 1 A

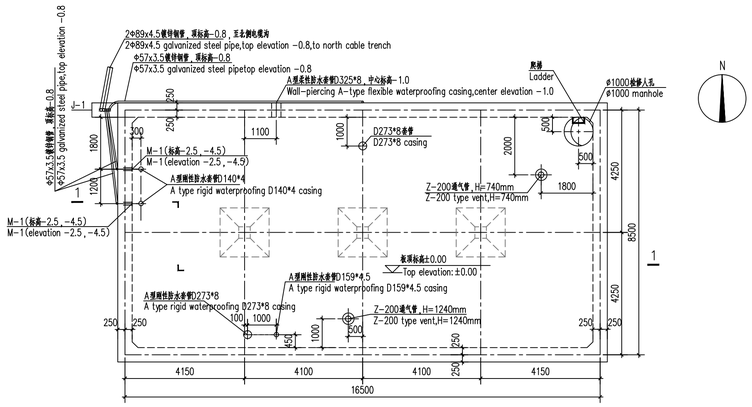
	姓名 Name	日期 Date
设计 Designed	付国峰	2019.11.12
校核 Checked	鲍伟	2019.11.12
审核 Reviewed	孟庆清	2019.11.12
批准 Approved	杨军	2019.11.12
比例 Scale	1:100	图号 DWG NO.

地下构筑物结构设计说明
Specification of structural design for underground structures

30-B003701S-T0509-001　版本 REV. A

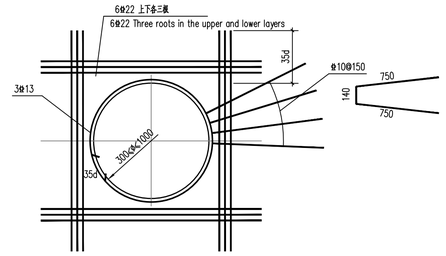
专业 Speciality	会签 Sign	日期 Date
水工	罗春	2019.11.12

594X841



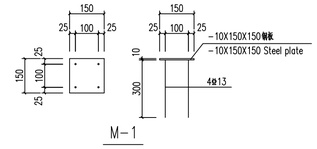
水池顶面布置图 1:100

Layout for top of pool

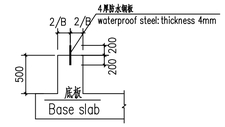


孔洞处加固图

Reinforcement details at hole

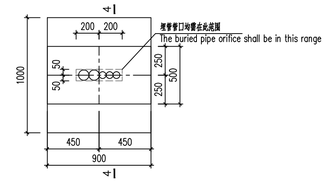


M-1

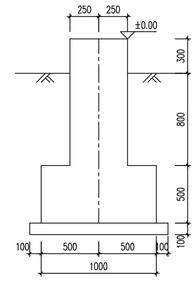


施工缝节点

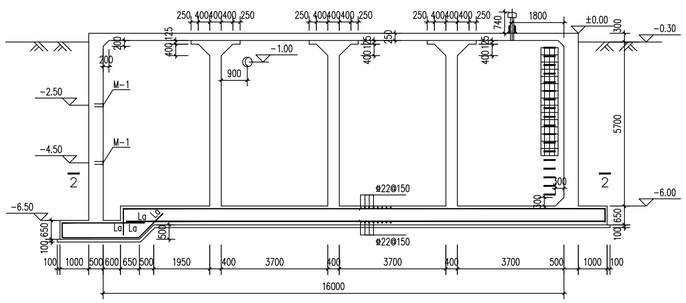
Construction joints



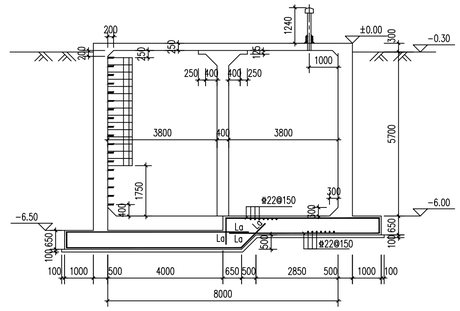
J-1



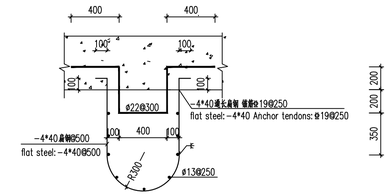
4-4



1-1 1:100

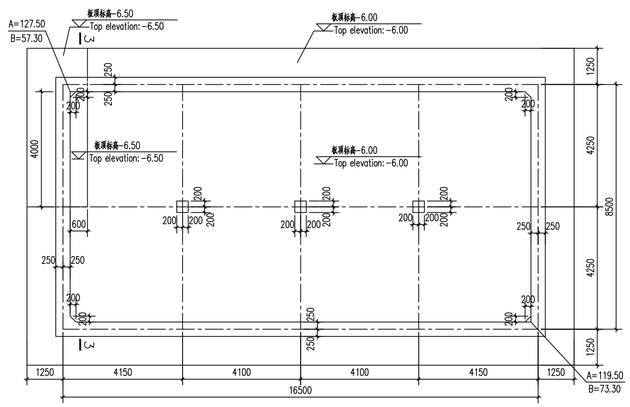


3-3 1:100



爬梯详图

Ladder details



2-2 1:100

说明：

1.场地标高为-0.30m，相当于绝对标高193.40m(EGM2008高程系统)。如无特别说明，本工程标高均采用相对标高。

2.土建施工时,请和工艺部分图纸结合,以免遗漏有关埋件。

3.人孔做法参考图集05S804-174。

4.埋管管口标高同基础表面标高，为±0.00。

5.J-1与水池池壁相交处设沉降缝，缝宽20mm，缝内填橡胶泡沫板和沥青麻丝。

6.各电缆埋管转弯半径不小于6倍管径，每根弯头不超过3个，直角弯不超过2个。

Specification：

1.The site level of -0.300 is equal to the absolute elevation of193.40(geodestic datum system EGM2008)
If not specified,the elevations of the project are adopt to the relative elevation.

2.In civil construction, please combine with the process drawings, so as not to miss the embedded parts.

3.Manhole uses insulation practices,refers to altas 05S804-174.

4. The elevation of buried pipe orifice is the same as that of foundation.

5.J-1 is provided with a settlement joint at the intersection of the pool wall. The width of the seam is 20mm.

6.The turning radius of each buried cable pipe shall not be less than 6 times of the pipe diameter, each elbow shall not be more than 3, and the right angle bend shall not be more than 2.

专业 Speciality	会签 Sign	日期 Date
电气一次	郑煜	2019.11.12
水工	甲瑶	2019.11.12

Reference Drawings

Drwg. No.	Title

Revisions

Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 State Grid 中国电力技术装备有限公司 CHINA ELECTRIC EQUIPMENT & TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

MLHVDC-LCS-ECI-DD-CS-0509-002A

	姓名 Name	日期 Date
设计 Designed	付国收	2019.11.12
校核 Checked	鸿伟	2019.11.12
审核 Reviewed	邵清清	2019.11.12
批准 Approved	杨岑	2019.11.12

水池平面图、剖面图

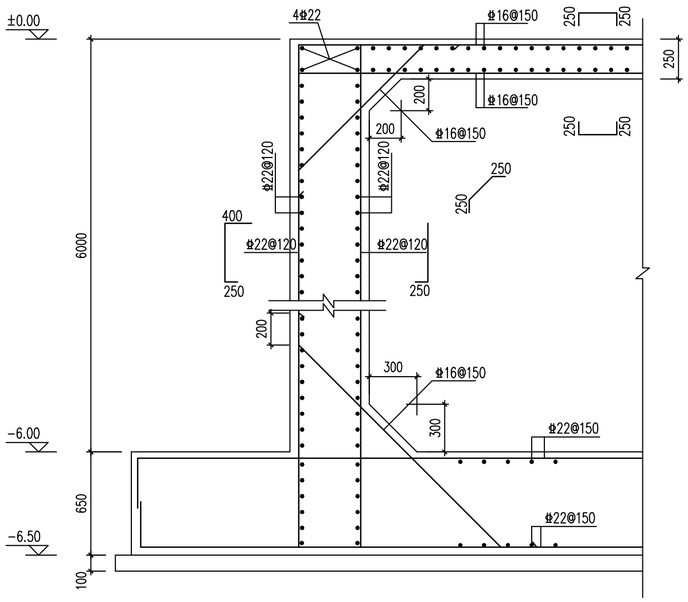
Plane and cross-section of pool

比例 Scale: 1:100

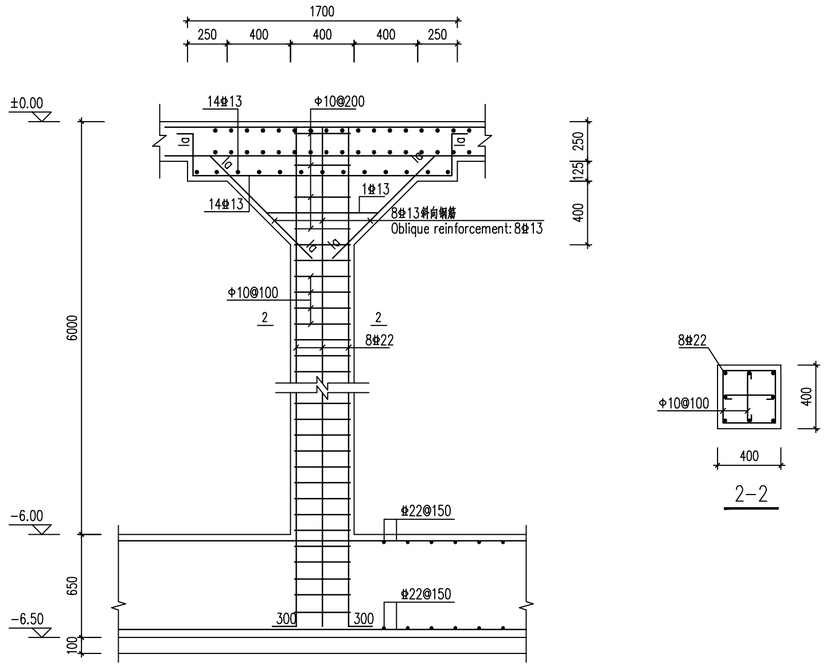
图号 DWG NO.: 30-B003701S-T0509-002

版本 REV.: A

594X841

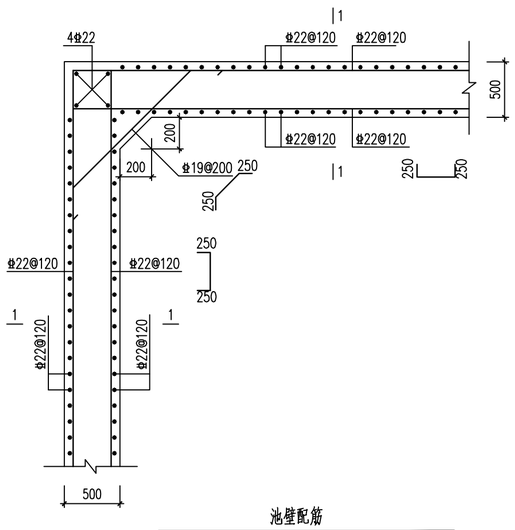


1-1



支柱及柱帽配筋

Column and caping reinforcement details



池壁配筋

Side wall reinforcement details

说明：

1.水池平面尺寸见图2。

2.池壁内钢筋间设置拉结筋Φ10@600。

Specification：

1.Industrial pool plane size refers to the Drwg.2.

2.Reinforcing bars Φ10@600 are arranged between the reinforcing bars in the wall of the pool wall.

Reference Drawings	
Drwg. No.	Title

Revisions

Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C I - D D - C S - 0 5 0 9 - 0 0 3 A

	姓名 Name	日期 Date
设计 Designed	付国顺	2019.11.12
校核 Checked	鸿伟	2019.11.12
审核 Reviewed	郑建清	2019.11.12
批准 Approved	杨少华	2019.11.12

水池配筋图

Pool reinforcement details

水工	甲楼	2019.11.12
专业 Speciality	会签 Sign	日期 Date

比例 Scale	1:25	图号 DWG NO.	30-B003701S-T0509-003	版本 REV.	A

420X594